

FORMULA, THEORY + PROOF / DERIVATIONS

Math 2123 | Section A + Section B | Set Distribution Edition

Purpose

A compact but detailed exam handbook containing the formulas you must recall, the theory you must be able to define/explain, and the proofs/derivations that repeatedly appear in the uploaded KUET Math 2123 material. Repetition-heavy items are placed first inside each set.

Credit: Made by 2223033

Source-grounded from uploaded lecture notes, KUET term questions, topic-wise solved compilations and reference books.

How to Use This Handbook

Set map and exam use

The handbook follows the same eight-set preparation map used in the question-bank analysis. Historical papers sometimes moved topics between sections, but the study map below keeps the preparation stable.

Set	Dominant topic	Repeat strength	Best use
A1	Matrix Fundamentals & Core Theorems	Very High	Definitions and short proofs
A2	Inverse, Rank, Consistency & Dependence	Very High	Mechanical row-operation logic
A3	Echelon/Normal Form & Eigen Block	Very High	Long structured matrix problems
A4	Vector Kinematics, Work & Cayley-Hamilton	Med-High	Formula-driven backup set
B5	Laplace Fundamentals & Direct Transform	Very High	Fast formula marks; safest anchor
B6	Inverse Laplace, Convolution, ODE & Beam	Very High	Long but highly structured
B7	Vector Differential Calculus	Very High	Short standardized methods
B8	Vector Integration	Very High	High marks; geometry and bounds matter

Section A: prepare A1-A3 first, keep A4 as a strong backup. **Section B:** B5 is the anchor; common routes are B5+B6+B7 or B5+B7+B8.

Difficult ideas are paired with short memory cues, while theorem names, formulas and exam notation remain in standard English.

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Master Formula Sprint

Section A - ultra-fast recall

Matrix + eigen + vector motion

$$A^T = A \quad \text{symmetric,}$$

$$A^T A = I \quad \text{orthogonal,}$$

$$A^2 = I \quad \text{involutory,}$$

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T),$$

$$\det A \neq 0 \iff \text{rank } A = n \iff A^{-1} \text{ exists,}$$

$$\det(A - \lambda I) = 0,$$

$$A = PDP^{-1}, p(\lambda) = \det(\lambda I - A)$$

$$r'(t) = v(t),$$

$$T = \frac{r'}{|r'|},$$

$$\kappa = \frac{|r' \times r''|}{|r'|^3},$$

$$W = \int_C F \cdot dr.$$

$$A^T = -A \quad \text{skew-symmetric,}$$

$$A^2 = A \quad \text{idempotent,}$$

$$A^k = 0 \quad \text{nilpotent for some } k \geq 1,$$

$$(A - \lambda I)x = 0,$$

$$\implies p(A) = 0,$$

$$r''(t) = a(t),$$

$$B = \frac{r' \times r''}{|r' \times r''|}, \quad N = B \times T,$$

$$\tau = \frac{(r' \times r'') \cdot r'''}{|r' \times r''|^2},$$

Section B - ultra-fast recall

Laplace + vector calculus + integration

$$\mathcal{L}\{f\}(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a},$$

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}, \quad \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2},$$

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a),$$

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s),$$

$$\mathcal{L}\{f'\} = sF - f(0), \quad \mathcal{L}\{f''\} = s^2 F - sf(0) - f'(0),$$

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{F(s)}{s},$$

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s),$$

$$\mathcal{L}\{f\}_{\text{period } T} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}},$$

$$\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(u)g(t-u) du,$$

$$\nabla\phi = (\phi_x, \phi_y, \phi_z), \quad \nabla \cdot F = P_x + Q_y + R_z,$$

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix},$$

$$D_{\hat{u}}\phi = \nabla\phi \cdot \hat{u}, \quad \max D_{\hat{u}}\phi = |\nabla\phi|,$$

$$\oint_C P dx + Q dy = \iint_R (Q_x - P_y) dA,$$

$$\oint_C F \cdot dr = \iint_S (\nabla \times F) \cdot n dS,$$

$$\iiint_{\partial V} F \cdot n dS = \iiint_V \nabla \cdot F dV.$$

Section A

Set A1 - Matrix Fundamentals & Core Theorems

VERY HIGH PRIORITY

QB: 2015-2025 recurring theory family

Matrix type master table

Type	Test / definition	Memory / immediate consequence
Square	$m = n$	same number of rows and columns
Diagonal	$a_{ij} = 0$ for $i \neq j$	only main diagonal may be nonzero
Scalar	diagonal with all diagonal entries equal	cI
Identity	scalar with diagonal entries 1	multiplicative identity
Upper triangular	$a_{ij} = 0$ for $i > j$	entries below diagonal are zero
Lower triangular	$a_{ij} = 0$ for $i < j$	entries above diagonal are zero
Symmetric	$A^T = A$	mirror across main diagonal
Skew-symmetric	$A^T = -A$	diagonal entries are zero
Orthogonal	$A^T A = A A^T = I$	$A^{-1} = A^T$
Singular	$\det A = 0$	inverse does not exist
Non-singular	$\det A \neq 0$	inverse exists
Idempotent	$A^2 = A$	$A^n = A$ for $n \geq 1$
Involutory	$A^2 = I$	$A^{-1} = A$
Nilpotent	$A^k = 0$ for some $k \geq 1$	all eigenvalues are zero
Hermitian	$A^\dagger = A$	complex analogue of symmetric
Skew-Hermitian	$A^\dagger = -A$	diagonal entries purely imaginary or zero
Elementary	obtained from I by one elementary operation	one-operation matrix
Equivalent	$B = PAQ$ with P, Q nonsingular	reachable by row and column operations
Trace	$\text{tr}(A) = \sum_i a_{ii}$	diagonal sum
Principal submatrix	retain same index set of rows and columns	diagonal subset preserved

Identity $\subset$ Scalar $\subset$ Diagonal. Each step adds a stronger condition.

Symmetric / skew decomposition

For every real square matrix,

$$A = P + Q, \quad P = \frac{1}{2}(A + A^T), \quad Q = \frac{1}{2}(A - A^T),$$

where $P^T = P$ and $Q^T = -Q$. For complex matrices, replace A^T by $A^\dagger$ to obtain Hermitian and skew-Hermitian parts.

Let $A = [a_{ij}]$ and $A^T = -A$. Elementwise, $a_{ji} = -a_{ij}$. Put $i = j$:

$$a_{ii} = -a_{ii} \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0.$$

Since i is arbitrary, every diagonal entry is zero.

If $A^T = A$ and A is nonsingular, then

$$(A^{-1})^T = (A^T)^{-1} = A^{-1}.$$

Hence A^{-1} is symmetric.

Define

$$P = \frac{1}{2}(A + A^T), \quad Q = \frac{1}{2}(A - A^T).$$

Then

$$P^T = \frac{1}{2}(A^T + A) = P, \quad Q^T = \frac{1}{2}(A^T - A) = -Q,$$

and $P + Q = A$.

For uniqueness, suppose $A = R + S$ where $R^T = R$ and $S^T = -S$. Then

$$A^T = R - S.$$

Adding and subtracting the equations,

$$R = \frac{1}{2}(A + A^T) = P, \quad S = \frac{1}{2}(A - A^T) = Q.$$

Thus the decomposition is unique.

Given $A^2 = I$, let $P = \frac{1}{2}(I + A)$. Then

$$P^2 = \frac{1}{4}(I + 2A + A^2) = \frac{1}{4}(2I + 2A) = \frac{1}{2}(I + A) = P.$$

Similarly, for $Q = \frac{1}{2}(I - A)$,

$$Q^2 = \frac{1}{4}(I - 2A + A^2) = \frac{1}{4}(I - A) = Q.$$

Hence both are idempotent.

Let the rows be (l_i, m_i, n_i) , $i = 1, 2, 3$, representing direction cosines of three mutually perpendicular unit directions. Then

$$l_i^2 + m_i^2 + n_i^2 = 1,$$

and for $i \neq j$,

$$l_i l_j + m_i m_j + n_i n_j = 0.$$

Therefore the (i, j) entry of AA^T is 1 when $i = j$ and 0 otherwise. Hence

$$AA^T = I,$$

so A is orthogonal.

In general $AB \neq BA$. Also $AB = 0$ does not force $A = 0$ or $B = 0$, and cancellation $AB = AC \Rightarrow B = C$ requires suitable invertibility assumptions.

Set A2 - Inverse, Rank, Consistency & Linear Dependence

VERY HIGH PRIORITY

Elementary operations and inverse

- Row swap: $R_i \leftrightarrow R_j$.
- Scaling: $R_i \rightarrow kR_i, k \neq 0$.
- Replacement: $R_i \rightarrow R_i + kR_j$.

Gauss-Jordan inverse:

$$[A \mid I] \xrightarrow{\text{same row operations}} [I \mid A^{-1}],$$

possible exactly when A is nonsingular.

D-R-I equivalence

For an $n \times n$ matrix,

$$\det A \neq 0 \iff \text{rank } A = n \iff A^{-1} \text{ exists} \iff Ax = 0 \text{ has only the trivial solution.}$$

Rank

$\text{rank}(A)$ is the maximum number of linearly independent rows (equivalently columns). Under elementary row or column operations, rank is unchanged. In echelon form, rank equals the number of pivots / nonzero rows.

Rouche-Capelli consistency rule

For $Ax = b$ with n unknowns, let $r = \text{rank}(A)$ and $r_a = \text{rank}([A \mid b])$.

$$\begin{aligned} r \neq r_a &\Rightarrow \text{inconsistent (no solution),} \\ r = r_a = n &\Rightarrow \text{unique solution,} \\ r = r_a < n &\Rightarrow \text{infinitely many solutions.} \end{aligned}$$

Look for a relation among coefficient rows first. The same relation must also hold among the right-hand sides; otherwise the augmented rank increases.

Linear independence

$v_1, \dots, v_k$ are linearly independent if

$$c_1 v_1 + \dots + c_k v_k = 0$$

implies $c_1 = \dots = c_k = 0$. Put the vectors as columns of M : if $\text{rank}(M) = k$, they are independent; otherwise dependent.

Homogeneous system

$Ax = 0$ always has the trivial solution $x = 0$. A nontrivial solution exists iff there is a free variable:

$$\text{rank}(A) < n.$$

For square A , this is equivalent to $\det A = 0$.

Set A3 - Echelon / Normal Form & Eigenvalue Block

VERY HIGH PRIORITY

REF, RREF and normal form

REF: nonzero rows above zero rows, pivot positions move rightward, and entries below each pivot are zero.

RREF / row canonical form: REF plus every pivot is 1 and is the only nonzero entry in its column.

Normal form under equivalence:

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix},$$

using row and column operations; the identity-block order r is the rank.

Row operations correspond to left multiplication (PA). Column operations correspond to right multiplication (AQ). Do not reverse the sides.

Eigenvalue / eigenvector engine

$$Ax = \lambda x, \quad x \neq 0.$$

D-S-V algorithm:

1. **D**: solve $\det(A - \lambda I) = 0$.
2. **S**: obtain the eigenvalues λ .
3. **V**: for each λ , solve $(A - \lambda I)x = 0$ and choose a nonzero solution.

High-value eigen properties

- Sum of eigenvalues (with algebraic multiplicity): $\sum \lambda_i = \text{tr}(A)$.
- Product: $\prod \lambda_i = \det A$.
- Triangular matrix: eigenvalues are its diagonal entries.
- A and A^T have the same characteristic polynomial.
- If A is invertible, eigenvalues of A^{-1} are $1/\lambda_i$.
- If $Ax = \lambda x$, then $A^m x = \lambda^m x$.
- Eigenvectors belonging to distinct eigenvalues are linearly independent.
- For real symmetric A , eigenvalues are real; eigenvectors for distinct eigenvalues are orthogonal.

Diagonalization

$$A = PDP^{-1} \quad \Longleftrightarrow \quad P^{-1}AP = D,$$

where the columns of P are linearly independent eigenvectors and the corresponding eigenvalues appear in the same order on the diagonal of D . An $n \times n$ matrix is diagonalizable if it has n linearly independent eigenvectors.

Set A4 - Vector Kinematics, Work & Cayley-Hamilton**MEDIUM-HIGH PRIORITY / STRONG BACKUP****Position - velocity - acceleration**

$$r(t) = x(t)i + y(t)j + z(t)k, \quad v = r'(t), \quad |v| = \text{speed}, \quad a = r''(t).$$

Acceleration decomposition:

$$a = a_T T + a_N N, \quad a_T = \frac{d|v|}{dt}, \quad a_N = \kappa |v|^2.$$

Frenet frame, curvature and torsion

$$T = \frac{r'}{|r'|}, \quad B = \frac{r' \times r''}{|r' \times r''|}, \quad N = B \times T,$$

$$\kappa = \frac{|r' \times r''|}{|r'|^3}, \quad \tau = \frac{(r' \times r'') \cdot r'''}{|r' \times r''|^2}.$$

At r_0 :

normal plane: $T \cdot (r - r_0) = 0$, osculating plane: $B \cdot (r - r_0) = 0$, rectifying plane: $N \cdot (r - r_0) = 0$.

Geometric meaning

T is tangent to the curve, N points toward local bending, and $B = T \times N$ completes the orthonormal frame. Curvature measures bending; torsion measures twisting out of the osculating plane.

Work done

For $F = Pi + Qj + Rk$,

$$W = \int_C F \cdot dr = \int_C (P dx + Q dy + R dz).$$

For a straight line from A to B , use

$$r(t) = A + t(B - A), \quad 0 \leq t \leq 1.$$

For a piecewise path, evaluate each segment separately and add.

The uploaded theory compilation asks for the identity

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B).$$

For completeness, a standard component derivation is included here. Using the Levi-Civita symbol,

$$[A \times (B \times C)]_i = \varepsilon_{ijk} A_j \varepsilon_{klm} B_l C_m.$$

Since

$$\varepsilon_{ijk} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl},$$

we obtain

$$[A \times (B \times C)]_i = B_i(A_j C_j) - C_i(A_j B_j),$$

which is exactly

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B).$$

Now cyclically permute A, B, C and add the three identities. Every term cancels because the dot product is commutative, giving

$$A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0.$$

Cayley-Hamilton theorem and inverse trick

If

$$p(\lambda) = \det(\lambda I - A),$$

then every square matrix satisfies

$$p(A) = 0.$$

If the constant term is nonzero (so A is invertible), multiply the matrix identity by A^{-1} and isolate A^{-1} . The same identity reduces high powers A^n to lower powers.

Section B

Set B5 - Laplace Fundamentals & Direct Transform

VERY HIGH PRIORITY / SAFEST ANCHOR

Definition and sufficient existence conditions

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

A standard sufficient condition for existence is: f is piecewise continuous on every finite interval and is of exponential order α , i.e. $|f(t)| \leq Me^{\alpha t}$ for all sufficiently large t . These conditions are sufficient, not necessary.

$f(t)$	$\mathcal{L}\{f(t)\}$	Cue
1	$1/s$	constant
t^n	$n!/s^{n+1}$	power $\rightarrow$ factorial
e^{at}	$1/(s-a)$	shift of s
$\sin at$	$a/(s^2 + a^2)$	sine numerator a
$\cos at$	$s/(s^2 + a^2)$	cosine numerator s
$\sinh at$	$a/(s^2 - a^2)$	hyperbolic: minus in denominator
$\cosh at$	$s/(s^2 - a^2)$	hyperbolic: minus in denominator
$u(t-a)$	e^{-as}/s	delayed switch

Core properties

If $\mathcal{L}\{f\} = F(s)$:

$$\begin{aligned} \mathcal{L}\{af + bg\} &= aF + bG, \\ \mathcal{L}\{e^{at}f(t)\} &= F(s-a), \\ \mathcal{L}\{f(at)\} &= \frac{1}{a}F\left(\frac{s}{a}\right), \quad a > 0, \\ \mathcal{L}\{t^n f(t)\} &= (-1)^n \frac{d^n F}{ds^n}, \\ \mathcal{L}\{f'(t)\} &= sF(s) - f(0), \\ \mathcal{L}\{f''(t)\} &= s^2F(s) - sf(0) - f'(0), \\ \mathcal{L}\left\{\int_0^t f(u)du\right\} &= \frac{F(s)}{s}, \\ \mathcal{L}\left\{\frac{f(t)}{t}\right\} &= \int_s^{\infty} F(u)du \quad (\text{when the required convergence holds}). \end{aligned}$$

Unit step and second shift

$$u(t-a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a, \end{cases} \quad \mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}.$$

If $\mathcal{L}\{f(t)\} = F(s)$,

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s).$$

More generally,

$$\mathcal{L}\{g(t)u(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}.$$

Start with the first expression, then at each switch point add

$$(\text{new expression} - \text{old expression})u(t-a).$$

This is the safest construction rule for converting a piecewise function into unit-step form.

Periodic transform

If $f(t+T) = f(t)$,

$$\mathcal{L}\{f(t)\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}.$$

Integrate only one full period; if it is piecewise, split only inside $[0, T]$.

$$\mathcal{L}\{c_1 f + c_2 g\} = \int_0^\infty e^{-st} [c_1 f(t) + c_2 g(t)] dt = c_1 F(s) + c_2 G(s).$$

If $F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\{e^{at} f(t)\} = \int_0^\infty e^{-st} e^{at} f(t) dt = \int_0^\infty e^{-(s-a)t} f(t) dt = F(s-a).$$

Assuming the boundary term vanishes at infinity,

$$\begin{aligned} \mathcal{L}\{f'(t)\} &= \int_0^\infty e^{-st} f'(t) dt \\ &= [e^{-st} f(t)]_0^\infty + s \int_0^\infty e^{-st} f(t) dt \\ &= sF(s) - f(0). \end{aligned}$$

Repeated application gives

$$\mathcal{L}\{f^{(n)}\} = s^n F - s^{n-1} f(0) - \dots - f^{(n-1)}(0).$$

Let

$$g(t) = \int_0^t f(u) du.$$

Then $g'(t) = f(t)$ and $g(0) = 0$. Taking transforms of g' ,

$$F(s) = sG(s) - 0 \Rightarrow G(s) = \frac{F(s)}{s}.$$

Therefore

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{F(s)}{s}.$$

Differentiate the defining integral with respect to s (under the usual validity conditions):

$$F'(s) = \frac{d}{ds} \int_0^\infty e^{-st} f(t) dt = \int_0^\infty (-t) e^{-st} f(t) dt.$$

Hence

$$\mathcal{L}\{t f(t)\} = -F'(s),$$

and repeated differentiation yields

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s).$$

Let $G(s) = \mathcal{L}\{f(t)/t\}$. Then, using the multiplication property on $g(t) = f(t)/t$,

$$F(s) = \mathcal{L}\{tg(t)\} = -G'(s).$$

Thus $G'(s) = -F(s)$. If $G(s) \rightarrow 0$ as $s \rightarrow \infty$, integration from s to ∞ gives

$$G(s) = \int_s^\infty F(u)du.$$

So

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u)du,$$

provided the relevant improper integrals converge.

Split the defining integral into periods:

$$\int_0^\infty e^{-st} f(t) dt = \sum_{n=0}^\infty \int_{nT}^{(n+1)T} e^{-st} f(t) dt.$$

Put $t = u + nT$ and use $f(u + nT) = f(u)$:

$$= \sum_{n=0}^\infty e^{-snT} \int_0^T e^{-su} f(u) du.$$

The geometric series sums to $1/(1 - e^{-sT})$, hence

$$\mathcal{L}\{f\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}.$$

Near $t = 0$,

$$\frac{e^{-st} \cos(at)}{t} \sim \frac{1}{t}.$$

Therefore the defining integral contains the divergent improper behavior

$$\int_0^\varepsilon \frac{dt}{t} = \infty.$$

Hence the Laplace transform of $\cos(at)/t$ does not exist as an ordinary improper Laplace integral.

High-priority transform

For $a > 0$ and $s > 0$,

$$\mathcal{L}\left\{\frac{\sin at}{t}\right\} = \int_s^\infty \frac{a}{u^2 + a^2} du = \tan^{-1}\left(\frac{a}{s}\right).$$

First derive the Dirichlet integral by Laplace regularization. Define

$$J(a, s) = \int_0^\infty e^{-st} \frac{\sin(at)}{t} dt, \quad s > 0.$$

Differentiate with respect to a :

$$\frac{\partial J}{\partial a} = \int_0^\infty e^{-st} \cos(at) dt = \frac{s}{s^2 + a^2}.$$

Since $J(0, s) = 0$,

$$J(a, s) = \int_0^a \frac{s}{s^2 + u^2} du = \tan^{-1}\left(\frac{a}{s}\right).$$

Let $s \rightarrow 0^+$: for $a > 0$,

$$\int_0^\infty \frac{\sin(at)}{t} dt = \frac{\pi}{2}.$$

Now integrate by parts:

$$I = \int_0^\infty \frac{\sin^2 t}{t^2} dt = \int_0^\infty \frac{\sin 2t}{t} dt = \frac{\pi}{2}.$$

(The boundary term vanishes at 0 and ∞ .)

Set B6 - Inverse Laplace, Convolution, ODE & Beam

VERY HIGH PRIORITY

Inverse Laplace method selector

Pattern	Best first method	Recognition clue
factorable rational function	partial fractions	linear / repeated / quadratic factors
quadratic in shifted s	complete square + first shift	$(s - a)^2 + b^2$
simple distinct denominator roots	Heaviside expansion	residue $P(a_k)/Q'(a_k)$
product of two known transforms	convolution	$F(s)G(s)$
$e^{-as}F(s)$	second shift / unit step	time delay a

Inverse patterns

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} &= e^{at}, \\ \mathcal{L}^{-1}\left\{\frac{s-a}{(s-a)^2+b^2}\right\} &= e^{at} \cos bt, \\ \mathcal{L}^{-1}\left\{\frac{b}{(s-a)^2+b^2}\right\} &= e^{at} \sin bt, \\ \mathcal{L}^{-1}\left\{\frac{1}{(s-a)^2-b^2}\right\} &= \frac{1}{b} e^{at} \sinh bt.\end{aligned}$$

For distinct simple roots a_k of Q ,

$$\frac{P(s)}{Q(s)} = \sum_k \frac{P(a_k)}{Q'(a_k)} \frac{1}{s - a_k}.$$

For

$$(f * g)(t) = \int_0^t f(u)g(t-u)du,$$

compute

$$\mathcal{L}\{f * g\} = \int_0^\infty e^{-st} \int_0^t f(u)g(t-u)du dt.$$

Interchange the order of integration over $0 \leq u \leq t < \infty$, and put $v = t - u$:

$$\begin{aligned}\mathcal{L}\{f * g\} &= \int_0^\infty f(u)e^{-su} \left[\int_0^\infty e^{-sv} g(v) dv \right] du \\ &= F(s)G(s).\end{aligned}$$

Therefore

$$\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(u)g(t-u)du.$$

ODE by Laplace - repeatable workflow

For an IVP:

1. Set $Y(s) = \mathcal{L}\{y(t)\}$.
2. Transform each derivative and insert initial conditions immediately.
3. Collect $Y(s)$ and solve algebraically.
4. Invert with partial fractions / shifts / convolution.
5. Check the initial conditions in the final answer.

Beam deflection

Using the sign convention of the uploaded beam compilation,

$$EI y^{(4)}(x) = w(x).$$

Boundary conditions:

End type	Conditions	Physical meaning
Clamped / embedded	$y = 0, y' = 0$	no deflection, no slope
Simply supported / hinged	$y = 0, y'' = 0$	no deflection, no bending moment
Free	$y'' = 0, y''' = 0$	no moment, no shear

Clamp locks deflection and slope. A hinge locks deflection but allows rotation, so $y' = 0$. A free end has zero bending moment and zero shear, so $y'' = y''' = 0$.

Discontinuous UDL and continuous simply-supported UDL

If a UDL w_0 acts only on $0 < x < a$,

$$w(x) = w_0[1 - u(x - a)].$$

For a simply supported beam of length L carrying constant w_0 and satisfying $EI y^{(4)} = w_0$,

$$y(x) = \frac{w_0}{24EI} (x^4 - 2Lx^3 + L^3x).$$

If downward load is defined negative in a different sign convention, the overall sign reverses.

Set B7 - Vector Differential Calculus

VERY HIGH PRIORITY

Nabla master table

For $F = Pi + Qj + Rk$,

$$\nabla = i\partial_x + j\partial_y + k\partial_z.$$

$$\nabla\phi = (\phi_x, \phi_y, \phi_z), \quad \nabla \cdot F = P_x + Q_y + R_z,$$

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix}, \quad \nabla^2 \phi = \phi_{xx} + \phi_{yy} + \phi_{zz}.$$

Physical meanings (2015 Q5c)

Gradient: points in the direction of greatest increase of a scalar field; its magnitude is the maximum directional rate. It is normal to a level surface $\phi = C$.

Divergence: measures net outward flux per unit volume; positive means source-like, negative sink-like, zero means solenoidal.

Curl: measures local rotational tendency; zero curl means irrotational.

Directional derivative and maximum rate

For direction d , first normalize:

$$\hat{u} = \frac{d}{|d|}.$$

Then

$$D_{\hat{u}} \phi = \nabla \phi \cdot \hat{u}, \quad \max_{|\hat{u}|=1} D_{\hat{u}} \phi = |\nabla \phi|,$$

and the maximizing direction is $\nabla \phi / |\nabla \phi|$ when $\nabla \phi \neq 0$.

For any unit vector $\hat{u}$,

$$D_{\hat{u}} \phi = \nabla \phi \cdot \hat{u}.$$

By Cauchy-Schwarz,

$$\nabla \phi \cdot \hat{u} \leq |\nabla \phi| |\hat{u}| = |\nabla \phi|.$$

Equality holds when $\hat{u}$ is parallel to $\nabla \phi$. Therefore the maximum directional derivative is $|\nabla \phi|$ in the gradient direction.

Tangent plane and normal line

For a level surface $F(x, y, z) = 0$ at $P = (x_0, y_0, z_0)$,

$$\nabla F(P) \cdot [(x, y, z) - P] = 0$$

is the tangent plane. A normal line is

$$\frac{x - x_0}{F_x(P)} = \frac{y - y_0}{F_y(P)} = \frac{z - z_0}{F_z(P)}$$

when the denominators are nonzero.

Along any differentiable curve $r(t)$ lying on $F(x, y, z) = C$,

$$F(r(t)) = C.$$

Differentiate:

$$\nabla F(r(t)) \cdot r'(t) = 0.$$

Thus ∇F is perpendicular to every tangent direction $r'(t)$ on the surface, so it is a normal vector.

Angle between surfaces

At a common point of $F = 0$ and $G = 0$,

$$\cos \theta = \frac{|\nabla F \cdot \nabla G|}{|\nabla F| |\nabla G|}.$$

The surfaces are orthogonal iff $\nabla F \cdot \nabla G = 0$.

Solenoidal, irrotational, conservative

$$\text{solenoidal} \iff \nabla \cdot F = 0, \quad \text{irrotational} \iff \nabla \times F = 0,$$

$$\text{conservative} \iff F = \nabla \phi.$$

On a simply connected domain (with suitable smoothness), $\nabla \times F = 0$ implies F is conservative.

High-value vector identities

$$\begin{aligned} \nabla(fg) &= f\nabla g + g\nabla f, \\ \nabla \cdot (fA) &= \nabla f \cdot A + f(\nabla \cdot A), \\ \nabla \times (fA) &= \nabla f \times A + f(\nabla \times A), \\ \nabla \times (\nabla \phi) &= 0, \\ \nabla \cdot (\nabla \times A) &= 0, \\ \nabla \cdot (\nabla \phi) &= \nabla^2 \phi. \end{aligned}$$

Set B8 - Vector Integration: Line, Surface & Volume

VERY HIGH PRIORITY / GEOMETRY FIRST

Integral type map

$$\begin{aligned} \text{work / circulation: } & \int_C F \cdot dr, \\ \text{scalar line integral: } & \int_C f ds, \\ \text{surface flux: } & \iint_S F \cdot n dS, \\ \text{volume total: } & \iiint_V f dV. \end{aligned}$$

Line integral algorithm

If $r = r(t)$, $a \leq t \leq b$,

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt.$$

Orientation is built into the parameter direction. Reverse the direction and the line integral changes sign.

Green's theorem - planar closed curve

For a positively oriented (counter-clockwise) simple closed curve C bounding R ,

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Clockwise orientation reverses the sign.

Useful vector area elements

For $z = g(x, y)$ with upward orientation,

$$n \, dS = (-g_x, -g_y, 1) \, dx \, dy.$$

For $x = g(y, z)$ with $+x$ orientation,

$$n \, dS = (1, -g_y, -g_z) \, dy \, dz.$$

For an implicit surface $G(x, y, z) = 0$, the normal direction is ∇G ; choose the sign to match the required orientation.

Stokes' theorem

If $C = \partial S$ and the orientations follow the right-hand rule,

$$\oint_C F \cdot dr = \iint_S (\nabla \times F) \cdot n \, dS.$$

Use it when a line integral around the boundary of a surface is easier to replace by a surface integral of curl, or vice versa.

Gauss divergence theorem

For a closed surface ∂V with outward normal,

$$\iint_{\partial V} F \cdot n \, dS = \iiint_V (\nabla \cdot F) \, dV.$$

Use it when a closed-surface flux is easier to replace by one volume integral of divergence.

Green = flat closed boundary in the plane. **Stokes** = boundary curve of a surface. **Gauss** = closed surface enclosing a volume.

Coordinate systems and Jacobians

Cartesian: $dV = dx \, dy \, dz$.

Cylindrical:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad dV = r \, dr \, d\theta \, dz.$$

Spherical:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi, \\ dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

Theory Question Bank - What You Should Be Able to Write

Section A theory checklist

1. Define with example: equivalent, singular, nonsingular, trace, symmetric, skew-symmetric, scalar, diagonal, lower/upper triangular, Hermitian, orthogonal, idempotent, involutory, nilpotent, elementary, principal submatrix.
2. Explain the difference between symmetric and skew-symmetric matrices.
3. State the symmetric + skew-symmetric decomposition formula.
4. Define rank; explain how rank is obtained from echelon form.
5. Define consistent/inconsistent systems; state the rank criteria for no/unique/infinite solutions.
6. Define trivial and nontrivial solutions of a homogeneous system.
7. Define linear independence and dependence.
8. Define REF, RREF / row canonical form, and normal form.
9. Define eigenvalue and eigenvector; list core eigen properties.
10. State the condition for diagonalizability.
11. Define velocity, acceleration, unit tangent, principal normal, binormal, curvature and torsion.
12. Explain normal, osculating and rectifying planes.
13. Define work done by a force field.
14. State the Cayley-Hamilton theorem and its uses.

Section B theory checklist

1. Define integral transform and identify the kernel of the Laplace transform.
2. Define the Laplace transform and state sufficient conditions for existence.
3. Define the unit step / Heaviside function.
4. State first and second shifting theorems.
5. State the periodic-function Laplace formula.
6. State the multiplication-by- t , derivative, integral and division-by- t properties.
7. Define inverse Laplace transform and convolution.
8. State the convolution theorem.
9. Explain the Laplace workflow for an IVP.
10. State the beam equation used in the notes and the clamped/hinged/free boundary conditions.
11. Define gradient, divergence, curl, Laplacian and directional derivative.
12. Explain the physical meaning of gradient and divergence; explain curl as local rotation.
13. Define solenoidal, irrotational and conservative fields.
14. State the angle formula for two surfaces and the tangent-plane formula.
15. Define line, surface and volume integrals.
16. State Green's, Stokes' and Gauss' theorems, including orientation requirements.

Proof / Derivation Priority Bank

Rank	Proof / derivation	Exam value	One-line memory route
1	$A = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$ and uniqueness	Very High	transpose; then add/subtract
2	Diagonal of skew-symmetric matrix is zero	Very High	$a_{ii} = -a_{ii}$
3	Symmetric nonsingular $A \Rightarrow A^{-1}$ symmetric	High	$(A^{-1})^T = (A^T)^{-1}$
4	Involutory $A \Rightarrow \frac{1}{2}(I \pm A)$ idempotent	High	expand square, use $A^2 = I$
5	Direction-cosine matrix is orthogonal	High	rows are orthonormal
6	Vector triple product + Jacobi identity	High	component identity, cyclic sum
7	Laplace linearity	High	distribute integral
8	First shift	High	combine exponentials
9	$\mathcal{L}\{f'\} = sF - f(0)$	Very High	integration by parts
10	Integral property $F(s)/s$	High	set $g = \int_0^t f$ and transform g'
11	Multiplication by t	High	differentiate $F(s)$
12	Division by t	High	let $g = f/t$, then integrate $G' = -F$
13	Periodic transform	Very High	split into periods + geometric series
14	Convolution theorem	Very High	double integral + change order
15	$\mathcal{L}\{\cos at/t\}$ does not exist	Repeated	near 0, integrand $\sim 1/t$
16	$\int_0^\infty \sin^2 t/t^2 dt = \pi/2$	Repeated	Dirichlet integral via Laplace
17	Maximum directional derivative	High	Cauchy-Schwarz
18	Gradient normal to a level surface	High	differentiate $F(r(t)) = C$

The uploaded theory compilation's vector-triple-product item states the triple-product identity and then uses it to establish the cyclic/Jacobi identity. This handbook expands that item with a standard component derivation so that the proof is complete rather than circular.

Last-Minute Error Shield

- **Matrix:** equivalent and row-equivalent are not identical ideas; full equivalence may use both row and column operations.
- **Normal form:** RREF and the equivalence normal form $[I_r \ 0 \ 0]$ are different objects.
- **Eigen:** $x = 0$ is never an eigenvector; keep the column order of P matched to the diagonal order of D .
- **Consistency:** compare $\text{rank}(A)$ with $\text{rank}([A \mid b])$ before deciding unique/infinite.
- **Laplace shift:** $F(s-a) \leftrightarrow e^{at}f(t)$; $F(s+a) \leftrightarrow e^{-at}f(t)$.
- **Second shift:** an e^{-as} factor must produce a delay and a unit-step factor in the inverse.
- **Convolution:** use $f(u)g(t-u)$ with $0 \leq u \leq t$.
- **Beam:** write the four boundary conditions before long algebra.
- **Directional derivative:** normalize the direction vector first.
- **Surface flux:** do not multiply by dS twice when using the vector area element $n dS$ directly.
- **Green/Stokes/Gauss:** check orientation and whether the curve/surface is actually closed.
- **Jacobians:** cylindrical contributes r ; spherical contributes $\rho^2 \sin \phi$.

Two-Page Final Revision Sheet

Section A - formulas to reproduce from memory

A1-A4

$$\begin{aligned}
 A^T &= A, \quad A^T = -A, \quad A^T A = I, \quad A^2 = A, \quad A^2 = I, \quad A^k = 0, \\
 A &= \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T), \\
 \det A \neq 0 &\iff \text{rank } A = n \iff A^{-1} \text{ exists}, \\
 r(A) \neq r([A|b]) &\Rightarrow \text{no solution}, \\
 r(A) = r([A|b]) = n &\Rightarrow \text{unique}, \quad r(A) = r([A|b]) < n \Rightarrow \text{infinite}, \\
 PAQ &= \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, \\
 \det(A - \lambda I) = 0, \quad (A - \lambda I)x &= 0, \quad A = PDP^{-1}, \\
 r' = v, \quad r'' = a, \quad T &= \frac{r'}{|r'|}, \quad B = \frac{r' \times r''}{|r' \times r''|}, \quad N = B \times T, \\
 \kappa &= \frac{|r' \times r''|}{|r'|^3}, \quad \tau = \frac{(r' \times r'') \cdot r'''}{|r' \times r''|^2}, \\
 W &= \int_C F \cdot dr, \quad p(\lambda) = \det(\lambda I - A) \Rightarrow p(A) = 0.
 \end{aligned}$$

Section B - formulas to reproduce from memory

B5-B8

$$\begin{aligned}
 \mathcal{L}\{f\} &= \int_0^\infty e^{-st} f(t) dt, \\
 1 &\leftrightarrow 1/s, \quad t^n \leftrightarrow n!/s^{n+1}, \quad e^{at} \leftrightarrow 1/(s-a), \\
 \sin at &\leftrightarrow a/(s^2 + a^2), \quad \cos at \leftrightarrow s/(s^2 + a^2), \\
 \mathcal{L}\{e^{at}f\} &= F(s-a), \quad \mathcal{L}\{t^n f\} = (-1)^n F^{(n)}(s), \\
 \mathcal{L}\{f'\} &= sF - f(0), \quad \mathcal{L}\{f''\} = s^2 F - sf(0) - f'(0), \\
 \mathcal{L}\left\{\int_0^t f\right\} &= F/s, \quad \mathcal{L}\{f/t\} = \int_s^\infty F(u) du, \\
 \mathcal{L}\{f(t-a)u(t-a)\} &= e^{-as} F(s), \\
 f(t+T) = f(t) &\Rightarrow \mathcal{L}\{f\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}, \\
 \mathcal{L}^{-1}\{FG\} &= \int_0^t f(u)g(t-u) du, \\
 EIy^{(4)} &= w(x), \\
 \nabla\phi &= (\phi_x, \phi_y, \phi_z), \quad \nabla \cdot F = P_x + Q_y + R_z, \\
 D_{\hat{u}}\phi &= \nabla\phi \cdot \hat{u}, \quad \max D = |\nabla\phi|, \\
 \oint_C Pdx + Qdy &= \iint_R (Q_x - P_y) dA, \\
 \oint_C F \cdot dr &= \iint_S (\nabla \times F) \cdot n dS, \\
 \iint_{\partial V} F \cdot n dS &= \iiint_V \nabla \cdot F dV, \\
 dV_{\text{cyl}} &= r dr d\theta dz, \quad dV_{\text{sph}} = \rho^2 \sin\phi d\rho d\phi d\theta.
 \end{aligned}$$

Source Basis and 60-Minute Revision Route

Source base used

This handbook was prepared from the uploaded Math 2123 material. The main source groups are:

- KUET Math 2123 term-final question collection covering 2015-2025;
- Matrix lecture PDFs covering matrix types, echelon/canonical/normal forms, consistency, inverse by elementary transformation, and eigenvalue/eigenvector;
- Laplace lecture PDFs and the uploaded Laplace question collection;
- Vector lecture notes and the complete vector-integration question compilation;
- The complete theory/proof compilation, unit-step guide, and beam-deflection/UDL compilation;
- Uploaded reference books including H.K. Dass, Schaum outlines, Shanti Narayan, and David C. Lay.

Historical year tags are used as exam-priority cues; topics are organized using the A1-A4 and B5-B8 preparation map.

1. 0-10 min: reproduce the two-page final formula sheet without looking.
2. 10-25 min: rehearse Proofs 1-5 (matrix block).
3. 25-45 min: rehearse Laplace derivations - derivative, multiplication by t , periodic formula, convolution, and the two repeated existence/integral proofs.
4. 45-55 min: revise gradient/divergence/curl, tangent plane, theorem selector, and beam boundary conditions.
5. 55-60 min: read the Last-Minute Error Shield only.

For a proof question, begin with the exact theorem/definition being used, show the key algebraic or integral step, and end with a clear concluding line. For a formula question, write the condition of validity whenever it matters (for example, simple roots in Heaviside expansion, periodicity $f(t+T) = f(t)$, or convergence for division by t).

Made by 2223033